PheW TuBe — High-Performance Streaming Platform

Front End Engineering - II (25CSE0203) — Project Based Evaluation-II (Phase 2)

Department of Computer Science & Engineering • Chitkara University

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Submitted To:

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Project Vision & Core Objectives

01. The Problem & Need

- High buffering latency and intrusive third-party ads in conventional web portals.
- Video viewing disruptions caused by unexpected external site redirects.
- Lack of accessible voice search options for hands-free query navigation.
- Cluttered interfaces without configurable playback settings (theme, autoplay, HD).
- Unrestricted route access causing data contamination without active user sessions.
- Need for a modular, high-speed streaming Single Page Application architecture.

02. What We Built (Phase 2 Capabilities)

- React 18 streaming platform with 100% in-site video playback (zero external redirecting).
- Native Web Speech API voice search microphone with real-time soundwave pulse overlay.
- Authentic YouTube settings panel: Dark/Light theme, Autoplay, HD 1080p, API live switch.
- Protected Route Gatekeeper locking Subscriptions, Playlists, Liked, and History for guests.
- Dynamic tabbed Auth Modal with instant credential validation and LocalStorage persistence.
- Continuous Infinite Scroll feed with resilient YouTube Data API v3 live query fallback.

03. Key Engineering Objectives

- Provide isolated, stutter-free in-app media rendering maintaining steady 60 FPS.
- Enforce strict client-side session boundaries with declarative route guards.
- Centralize global state across playlists, likes, and watch history via AppContext.
- Build resilient hybrid data architecture combining Google YouTube API v3 and offline mocks.
- Ensure fluid multi-device responsiveness across mobile, 1024px tablet, and wide desktop.

Syllabus-Aligned Technology Stack

React 18 (SPA Engine)

- Declarative component-driven UI architecture.
- Virtual DOM reconciliation & Fiber concurrency.
- Optimized state hooks (useState, useEffect, useRef).
- Strict mode lifecycle validation & fast reconciliation.

Vite 5 (Build Tool)

- Lightning-fast Hot Module Replacement (HMR).
- Native ES Module development server (1.16s build).
- Optimized Rollup production bundler & minification.
- Zero-config ultra-fast development environment.

React Router v6

- Client-side Single Page Application navigation.
- Dynamic URL parameters (/watch/:id, /channel/:id).
- Protected Route security wrapper (<ProtectedRoute />).
- Declarative Link and Navigate redirection components.

Centralized State Layer

- Global AppContext provider at App.jsx application root.
- Synchronizes Liked, Watch Later, and History states.
- Persistent session & theme storage in LocalStorage.
- Optimistic UI updates for immediate interactive clicks.

YouTube Data API v3

- RESTful Google Cloud YouTube Data API integration.
- Dynamic channel & video query search engine.
- Resilient offline curated mock dataset (youtubeData.js).
- In-app settings switch to toggle live cloud API.

Tailwind CSS & Web APIs

- Native Web Speech API for voice recognition.
- react-infinite-scroll-component dynamic pagination.
- Responsive grid (1024px iPad Mini & Mobile layouts).
- Glassmorphic modal containers & animated pulse overlays.

Core Component Architecture & Modularity

Navbar.jsx
- Real-time search bar with autocomplete suggestions.
- Native Web Speech API voice search mic trigger.
- User avatar profile menu, theme switch & notifications.
- Hamburger button toggling collapsible responsive sidebar.

WatchPage.jsx
- In-site video streaming player with zero redirecting.
- Interactive action bar (Like, Dislike, Share, Watch Later).
- Channel header with dynamic Subscribe counter.
- Interactive comments thread with useRef autofocus.

AuthModal.jsx
- Tabbed authentication modal (Sign In & Sign Up).
- Form validation checking existing registered profiles.
- Automatic guest login upon successful account creation.
- Session persistence storing active user in LocalStorage.

SettingsModal.jsx
- Authentic YouTube settings panel modal interface.
- Display Theme toggle (Dark Mode vs Light Mode).
- Autoplay Next Video & HD 1080p Stream toggles.
- YouTube Data API v3 live API key manager & sync.

VideoCard.jsx
- 16:9 thumbnail display with duration timestamp badge.
- Channel avatar, verified checkmark & view metrics.
- Hover elevation animation & direct watch navigation.
- Integrated into continuous infinite scroll feed grid.

Sidebar.jsx & ProtectedRoute
- Primary navigation (Home, Shorts, Subscriptions).
- Protected You section (History, Playlists, Liked).
- Lock screen redirecting guests to AuthModal on click.
- Fully responsive collapsible mobile drawer menu.

Featured Content Catalog & Streaming Matrix



React 18 Course

1.9M Views

TECH • EDUCATION

Chai aur Code

Complete React tutorial covering hooks, Fiber, and DOM.

Quality: 1080p HD • 8:32:05



India's Got Latent

67M Views

COMEDY • SPECIAL

Samay Raina

Standup comedy panel featuring unfiltered roasts.

Quality: 1080p HD • 52:26



$250M Island

355M Views

CHALLENGE • BEAST

MrBeast

Surviving 7 days on the world's most expensive island.

Quality: 4K Ultra HD • 15:24



DSA In-Depth

6.4M Views

CODING • COURSE

freeCodeCamp

Algorithms & data structures complete curriculum.

Quality: 1080p HD • 5:22:09



GTA V Stunts

12M Views

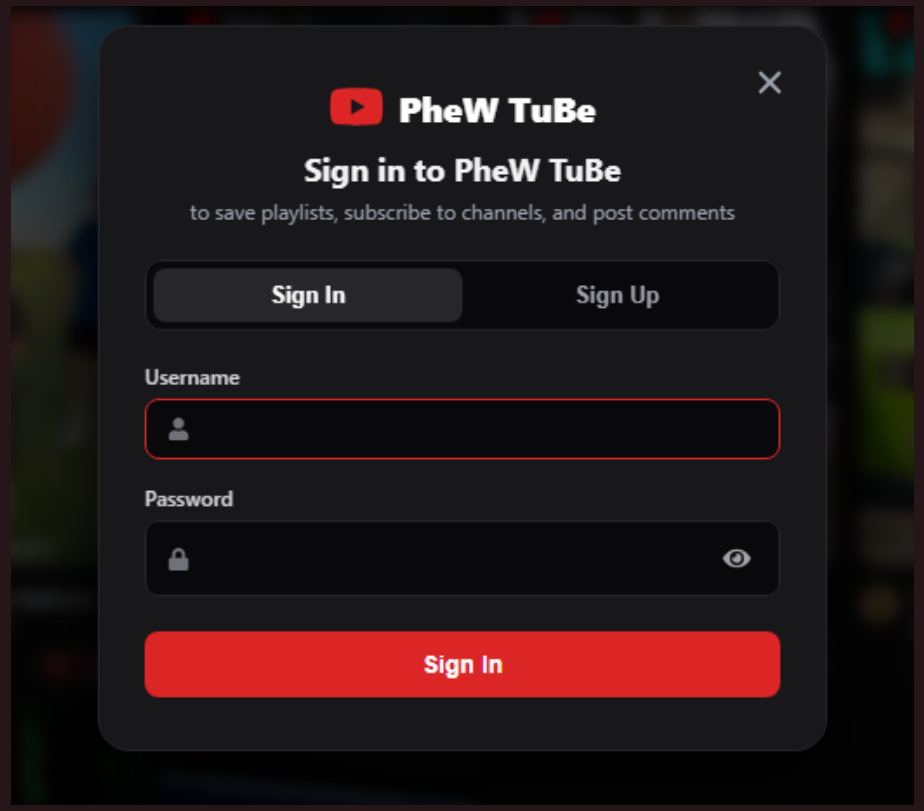
GAMING • POPULAR

Techno Gamerz

Insane mega ramp stunts across the GTA V map.

Quality: 1080p HD • 14:18

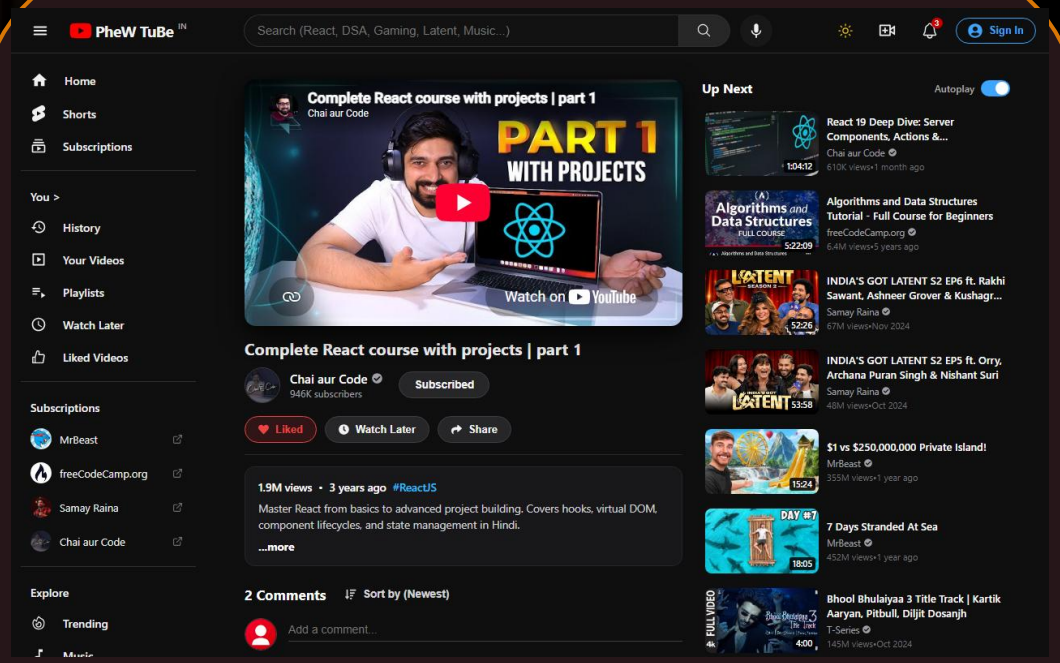
Guest Authentication & Protected Routing Gatekeeper



Authentication & Protected Route Architecture

- Dedicated Auth Modal: Tabbed interface switching between Sign In and Sign Up with username and password validation.
- Strict Protected Route Gatekeeper: Subscriptions, Playlists, Liked Videos, and History remain strictly locked until authenticated.
- Seamless Guest Onboarding: Attempting to access protected sections triggers Auth Modal without unmounting background player.
- Duplicate & Credentials Validation: Prevents duplicate usernames during sign up and prompts immediate sign in for existing users.
- Persistent Session Architecture: Stores active user session in LocalStorage, preserving authentication across page reloads.

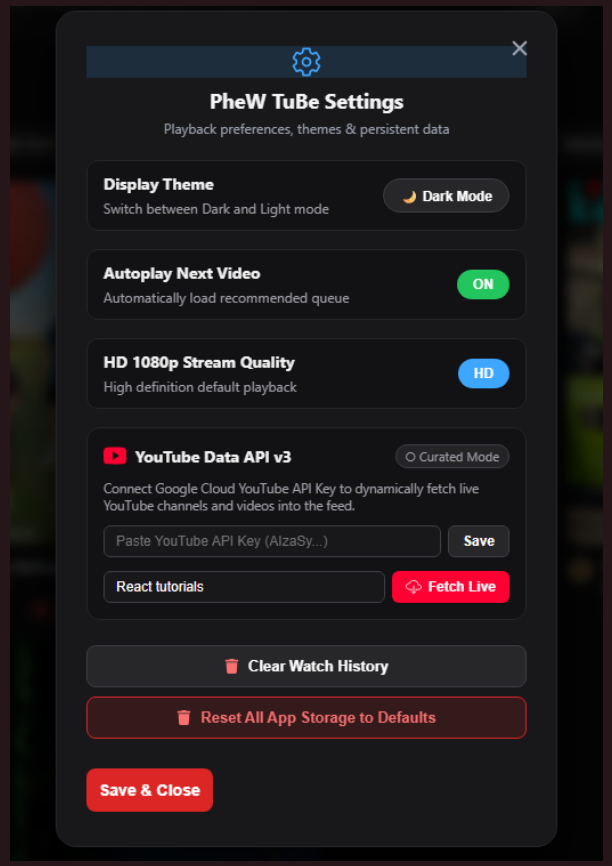
In-Site Video Player (No Redirecting) & Action Controls



In-Site Streaming & Interactive Controls

- 100% In-Site Video Playback: Videos stream directly on the /watch/:id route inside PheW TuBe with zero external redirects.
- Native Web Speech API Mic: Hands-free voice recognition with glowing soundwave pulse overlay and auto query search.
- Working Interactive Buttons: Functional Like toggle with counter increments, Share link copier, and Watch Later bookmarking.
- Channel Subscription Engine: Dynamic Subscribe/Subscribed toggle with subscriber count state synchronization.
- Real-Time Comment Discussion: Allows posting new comments with useRef auto-focus, timestamp sorting, and avatar updates.

Real YouTube Settings Panel & Infinite Scroll Engine



Settings Preferences & Continuous Pagination

- Authentic YouTube Settings UI: Replicates YouTube's genuine settings dialog with clean modal backdrop and smooth exit.
- Display Theme Engine: Instant switch between Dark Mode and Light Mode updating all backgrounds, cards, and text.
- Autoplay & HD Stream Toggles: Configurable Autoplay Next Video and default HD 1080p stream resolution preference.
- YouTube Data API v3 Live Switch: Toggle between offline curated mock library and Google Cloud live API with custom API key.
- Continuous Infinite Scroll: react-infinite-scroll-component automatically loads more video cards on scroll down without reload.

THANK YOU

PHEW TUBE — HIGH-PERFORMANCE STREAMING PLATFORM

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